

Drug reaction with eosinophilia and systemic symptoms (DRESS): clinicopathological study of 45 cases.

PMID: 26354734 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) is a rare and severe adverse drug reaction. Large detailed studies of histopathological features of DRESS are sparse and suggest an association between keratinocyte damage and the severity of visceral involvement. To describe the dermatopathological features in a large series of DRESS and their possible association with clinical features and the severity of the disease. A retrospective analysis of the clinicobiological and dermatopathological features in a monocentric cohort of patients with DRESS. From January 2005 to January 2013, 45 patients were validated as probable or definite cases of DRESS. The median age was 64 years (range 3-87). The most frequent clinical and biological features included: fever $\geq 38.5^{\circ}\text{C}$ (95%), facial oedema (72%), enlarged lymph nodes (51%), visceral involvement (75%), blood eosinophilia (97%) and atypical lymphocytes (82%). Severe DRESS occurred in 24% and a fatal outcome in 6% of patients. Histopathological analysis showed that no specific histopathological pattern was characteristic for DRESS. However, several changes in different cutaneous compartments were observed in 2 of 3 of cases. Spongiosis (55%) and keratinocyte damage (53%) were the most common epidermal changes. Spongiosis was associated with non-severe DRESS ($P = 0.041$) whereas confluent keratinocyte necrosis correlated with severe DRESS ($P = 0.011$). Vascular changes were frequent (88%). A moderate dermal perivascular lymphocytic infiltrate was invariably present, containing eosinophils, neutrophils and/or atypical lymphocytes in 57% of cases. Epidermal changes are indicative for the severity of DRESS.

Serious adverse effects of selected antiseizure medications used for treatment of focal onset seizures.

PMID: 39714228 | Indexed in PubMed

The third-generation antiseizure medications used for the treatment of focal seizures, lacosamide, eslicarbazepine acetate, perampanel, brivaracetam, and cenobamate, may elicit serious adverse reactions which could be preventable if a prescriber is acquainted with the risk factors. The literature search was conducted in MEDLINE, SCOPUS, and EBSCO databases, without time and language restrictions. Only clinical studies, observational human studies, case reports, and case series that reported serious adverse drug reactions and risk factors were considered. In order to prevent psychiatric side effects, the use of these drugs in patients with psychiatric disorders in the past should be avoided, or intensive monitoring of patients should be undertaken in order to detect psychosis or depression as early as possible. When prescribed eslicarbazepine and perampanel, serum sodium and ammonia levels should be periodically monitored. Lacosamide is associated with rhythm and conduction disturbances in the myocardium, so caution and ECG monitoring are required in people with cardiac diseases. Eslicarbazepine, cenobamate, and lacosamide can cause Drug Rash with Eosinophilia and Systemic Symptoms (DRESS) syndrome and other allergic manifestations, so they should be used with caution in people with allergies in medical history.

DRESS syndrome: Part I. Clinical perspectives.

PMID: 23602182 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome, also referred to as drug-induced hypersensitivity syndrome, is a distinct, potentially life-threatening adverse reaction. It is seen in children and adults most often as a morbilliform cutaneous eruption with fever, lymphadenopathy, hematologic abnormalities, and multiorgan manifestations. Historically, it was most frequently linked with phenytoin and known as phenytoin hypersensitivity syndrome. However, because many other medications were found to produce the same reaction, another name was in order. Anticonvulsants and sulfonamides are the most common offending agents. Its etiology has been linked with lymphocyte activation, drug metabolic enzyme defects, eosinophilia, and human herpesvirus-6 reactivation. DRESS has a later onset and longer duration than other drug reactions, with a latent period of 2 to 6 weeks. It may have significant multisystem involvement, including hematologic, hepatic, renal, pulmonary, cardiac, neurologic, gastrointestinal, and endocrine abnormalities. This syndrome has a 10% mortality rate, most commonly from fulminant hepatitis with hepatic necrosis.

[Drug reaction with eosinophilia and systemic symptoms (DRESS)].

PMID: 23424909 | Indexed in PubMed

DRESS (drug reaction with Eosinophilia and systemic symptoms) is a syndrome classically considered as a severe cutaneous drug adverse reaction. But visceral manifestations (renal, liver, lung, heart...) may be at the forefront. It presents clinically as an exanthema evolving to erythroderma with facial edema, associated with lymphadenopathy, high fever, visceral involvement (hepatitis, renal failure, pneumonitis, or hemophagocytic syndrome), eosinophilia preceded by lymphopenia and/or atypical lymphocytes. DRESS is characterized by a long delay between the first drug intake and its development (2 weeks to 3 months) and by its long course (more than 2 weeks) with flares even after drug discontinuation. Its pathophysiology is unique: it is the consequence of the immune response against Herpesvirus (HHV6, EBV, CMV) reactivation. It is induced by some drugs (allopurinol, anticonvulsants, sulfasalazine, minocycline...). Its early diagnosis is necessary for a rapid discontinuation of the culprit drug. Its management includes a long-term followup, and according to the severity either topical steroids, systemic steroids, intravenous gammaglobulins, or antiviral.

Drug Reaction with Eosinophilia and Systemic Symptoms: an update on pathogenesis.

PMID: 23062470 | Indexed in PubMed

The syndrome termed 'Drug Reaction with Eosinophilia and Systemic Symptoms' (DRESS) is an unpredictable, life-threatening condition associated with adverse reactions to therapy. Although the etiology of DRESS is poorly understood, genetic susceptibility markers have been identified within the HLA complex and there are several prevailing models of pathogenesis. Modification of host antigens by haptens (drugs or their metabolites), or non-covalent drug binding to endogenous proteins (the p-i concept), may drive pro-inflammatory immune responses in patients. Alternatively, a viral trigger for DRESS has been proposed based on the concomitant detection of herpesviruses and the recent demonstration of Epstein-Barr virus-specific immune responses in DRESS patients. In the present review, we discuss the latest findings concerning the pathogenesis of drug reactions and known risk factors for DRESS.

Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome.

PMID: 33341203 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is a severe cutaneous drug reaction characterized by fever, lymphadenopathy, hematologic abnormalities, multisystem involvement, and viral reactivation. Although most patients with DRESS syndrome are able to fully recover, a subset of patients go on to have a prolonged course with recurrence, and/or autoimmune complications. Severe systemic involvement is associated with significant morbidity and mortality. Viral reactivation, especially of human herpes virus 6, Epstein-Barr virus, and cytomegalovirus, is a common feature of DRESS, with a high viral load and antibody titers being associated with poor outcomes. Aside from prompt discontinuation of the offending drug, treatment for patients with significant disease consists of systemic therapy with corticosteroids.

The systemic treatments for drug reaction with eosinophilia and systemic symptoms (DRESS) beyond corticosteroids.

PMID: 39156598 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS), also known as drug-induced hypersensitivity syndrome (DiHS), is a severe type of cutaneous adverse reaction. The gold standard therapy for DRESS involves the discontinuation of the culprit drug, supportive therapies, and administration of corticosteroids. However, in cases of primary treatment failure or suboptimal response, there arises an urgent need for alternative interventions. This review focuses on exploring alternative systemic therapies for patients with steroid-resistant DRESS, steroid-dependent DRESS, or refractory DRESS, encompassing immunosuppressive agents, intravenous immunoglobulin, plasmapheresis, biologics, and small molecule drugs, with an emphasis on their clinical efficacy and the underlying mechanisms in the treatment of DRESS. Furthermore, this review provides a summary of potential management strategies and laboratory workup during the treatment of DRESS.

Hair loss after drug reaction with eosinophilia and systemic symptoms (DRESS): A multicentric retrospective case series.

PMID: 36651019 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) is a severe cutaneous drug adverse reaction characterized by various cutaneous and systemic manifestations. However, reports on the various patterns of alopecia after DRESS are lacking. Thus, we aimed to describe cases of alopecia after DRESS and review the literature. This multicentric retrospective study reviewed the records of 182 patients diagnosed with DRESS from 2009 to 2021; of these, 10 who had alopecia after DRESS were included. Patients were diagnosed with permanent alopecia (n = 4), telogen effluvium (n = 5), and alopecia areata (n = 1), and were treated with topical minoxidil or alfatradiol (6; 60%), topical corticosteroids (3; 30%), dietary supplements (6; 60%), systemic corticosteroids (1; 10%), and intralesional corticosteroid injection (2; 20%). Although patients with permanent alopecia did not show

hair regrowth after 6 months, those with telogen effluvium and alopecia areata experienced marked clinical improvement within 6 months. Various types of alopecia can persist over an extended period, even after the resolution of an acute episode of DRESS.

Vesiculopustular drug reaction with eosinophilia and systemic symptoms induced by levetiracetam.

PMID: 39104635 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) is a severe cutaneous adverse reaction characterised by fever, lymphadenopathy, morbilliform rash, haematologic abnormalities, and multiorgan involvement. Herein, we describe a 32-year-old female presenting with a 9-day history of facial oedema, cervical and inguinal lymphadenopathy, and a pruritic rash comprised of vesicles and pustules on her face, trunk, and extremities. Her only medications were valproate, which she had been taking for several years, and levetiracetam, which was initiated 41 days prior to rash onset. On the 16th day of her rash, she was diagnosed with DRESS induced by levetiracetam (Registry of Severe Cutaneous Adverse Reactions: 5). At this point, her absolute eosinophil count was 0.9×10^9 cells/L and aspartate and alanine transaminase levels were 357 and 339 U/L, respectively. Pustules with a morbilliform rash may occur in up to 30% of DRESS cases. In rarer instances, as in our patient, DRESS can present with isolated pustules and vesicles. Similarly, although rare, DRESS can be induced by levetiracetam.

Drug Reaction With Eosinophilia and Systemic Symptoms: A Systematic Review.

PMID: 36893848 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) is a potentially life-threatening drug reaction; recognizing the diversity of its clinical presentations, implicated drugs, and management modalities can aid in diagnosis and reduce morbidity and mortality. To review the clinical features, drug causes, and treatments deployed in DRESS. This review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to review publications relating to DRESS published between 1979 and 2021. Only publications with a RegiSCAR score of 4 or greater were included (indicating "probable" or "definite" DRESS). The PRISMA guidelines were used for data extraction and the Newcastle-Ottawa scale for quality assessment (Pierson DJ. *Respir Care* 2009;54:1372-8). The main outcomes included implicated drugs, patient demographics, clinical manifestations, treatment, and sequelae for each included publication. A total of 1124 publications were reviewed, and 131 met the inclusion criteria, amounting to 151 cases of DRESS. The most implicated drug classes were antibiotics, anticonvulsants, and anti-inflammatories, although up to 55 drugs were implicated. Cutaneous manifestations were present in 99% of cases, with a median onset of 24 days and maculopapular rash the most common morphology. Common systemic features were fever, eosinophilia, lymphadenopathy, and liver involvement. Facial edema was present in 67 cases (44%). Systemic corticosteroids were the mainstay of DRESS-specific treatment. A total of 13 cases (9%) resulted in mortality. DRESS diagnosis should be considered in the presence of a cutaneous eruption, fever, eosinophilia, liver involvement, and lymphadenopathy. The class of implicated drug may influence outcome, as allopurinol was associated with 23% of cases that resulted in death (3 cases). Given potential DRESS complications and mortality, it is important that DRESS is recognized early so that any

suspect drugs are ceased promptly.
